MI HDMI API

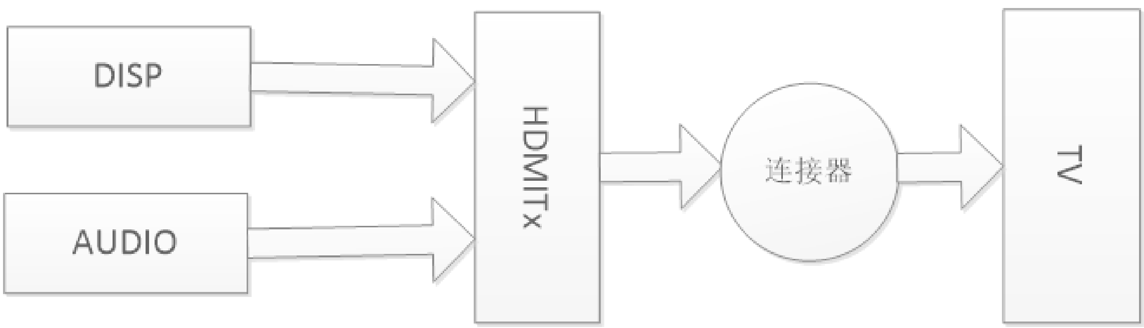
1 Overview

1.1. Module Description

HDMI (High Definition Multimedia Interface), namely high-definition multimedia interface, is a fully digital video and sound transmission interface, which can transmit uncompressed audio and video signals at the same time.

HDMI can be divided into `TX 端` and `RX端` , TX is the soure device, used to transmit data to RX (sink device). This module currently only supports HDMITx based on HDMI ver1.4.

1.2. Data flow block diagram



Picture 1-1

1.3. Features of each platform

Table 1-1

Platform Support Features	Taiyaki/Takoyaki
1920x1080@24	Y
1920x1080@25	Y
1920x1080@30	Y
1920x1080@50	Y
1920x1080@60	Y
1280x720@60	Y
1280x720@50	Y
720x576@50	Y
720x480@60	Y
1280x1024@60	Y
1024x768@60	Y
1440x900@60	Y
1600x1200@60	Y
1280x800@60	Y
1366x768@60	Y
1680x1050@60	Y
3840x2160@30	Y

Platform Support Features Taiyaki/Takoyaki

2560x1440@30	Y
Audio_32K	Y
Audio_48K	Y
1920x1080@59.94	N
1920x1080@29.97	N
1280x720@59.94	N
1280x720@29.97	N

1.4. Keyword Description

- EDID

  Extended display identification data

  Refers to the screen resolution information, including the manufacturer's name and serial number.

- InfoFrame

  The HDMI specification, which belongs to the auxiliary data category, is used to inform the sink device of various characteristics of the data to be transmitted.

- DeepColor

  That is, the dark mode is used to represent more accurate colors, and the amount of data will increase accordingly.

- SinkInfo

  Device information on the sink side.

2. API Reference

2.1. Function Module API

table 2-1

API name	Function
MI_HDMI_Init [#22-mi_hdmi_init]	Initialize HDMI
MI_HDMI_DeInit [#23-mi_hdmi_deinit]	deinitialize HDMI
MI_HDMI_Open [#24-mi_hdmi_open]	Open HDMI
MI_HDMI_Close [#25-mi_hdmi_close]	Turn off HDMI
MI_HDMI_SetAttr [#26-mi_hdmi_setattr]	Set HDMI properties
MI_HDMI_GetAttr [#27-mi_hdmi_getattr]	Get HDMI properties
MI_HDMI_Start [#28-mi_hdmi_start]	enable HDMI
MI_HDMI_Stop [#29-mi_hdmi_stop]	stop HDMI
MI_HDMI_GetSinkInfo [#210-mi_hdmi_getsinkinfo]	Get the capability information of HDMI sink
MI_HDMI_SetAvMute [#211-mi_hdmi_setavmute]	Set HDMI audio and video mute function
MI_HDMI_ForceGetEdid [#212-mi_hdmi_forcegetedid]	Get HDMI EDID information
MI_HDMI_SetDeepColor [#213-mi_hdmi_setdeepcolor]	Set HDMI Deep Color Mode
MI_HDMI_GetDeepColor [#214-mi_hdmi_getdeepcolor]	Get HDMI Deep Color Mode
MI_HDMI_SetInfoFrame [#215-mi_hdmi_setinfoframe]	Set HDMI Info Frame information

API name Function

MI_HDMI_GetInfoFrame [#216-mi_hdmi_getinfoframe]	Get the Info Frame information of the currently set HDMI
MI_HDMI_SetAnalogDrvCurrent [#217-mi_hdmi_setanalogdrvcurrent]	Set HDMI channel drive current
MI_HDMI_InitDev [#218-mi_hdmi_initdev]	Initialize HDMI device
MI_HDMI_DeInitDev [#219-mi_hdmi_deinitdev]	Deinitialize HDMI device

2.2. MI_HDMI_Init

- Function

  Initialize the HDMI module.

- grammar

```
MI_S32 MI_HDMI_Init(MI_HDMI_InitPara_t *pstHdmiPara);
```

- formal parameter

  Table 2-2

parameter name	describe	input Output
pstHdmiPara	Initialize the parameter structure pointer, which can be configured as NULL	enter

- return value

  0 Initialization succeeded.

  Non-zero initialization failed, please refer to the error code [#jump] for details .

- rely

  - Header files: mi_hdmi.h, mi_hdmi_datatype.h

  - Library file: libmi_hdmi.a

- Notice

  After HDMI registers the event callback, the system will notify the AP of hot swapping and other operations in the form of a callback. If the user does not need to make a special response to the corresponding event, the function pointer in the initialization parameter structure can be set to NULL.

- Special Instructions

  The pfnHdmiEventCallback callback registered in Init is mainly to call back the plug and unplug events to the App. The current design is to open a thread at the SDK layer, and then check the corresponding time and report it to the callback registrant

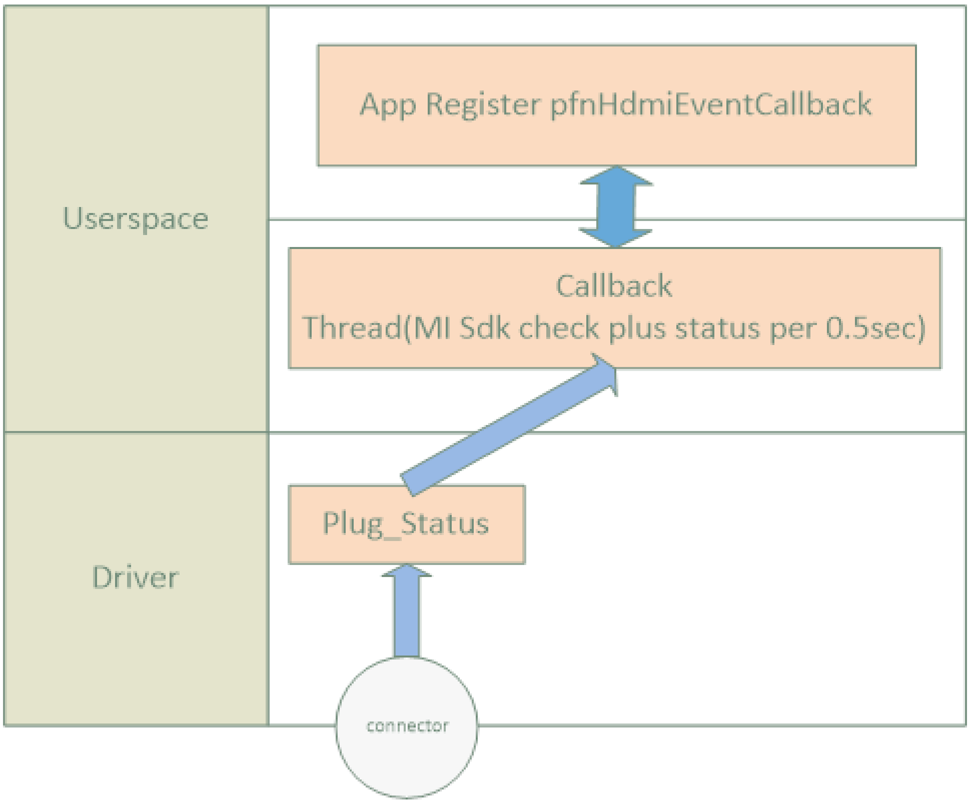


  Figure 2-1

- Example

```
1. MI_HDMI_InitPara_t stHdmiPara ; _
2. MI_HDMI_Attr_t stAttr ; _
3. MI_HDMI_TimingType_eTimingType ; _   _
```

```
MI_HDMI_Edid_t stEdid ; _
/   * Start DISP */
eTimingType = E_MI_HDMI_TIMING_1080_60P ; _
   /* Enable HDMI */
stHdmiPara .   pCallBackArgs = NULL ; _
stHdmiPara .   pfnHdmiEventCallback = NULL ; _
MI_HDMI_Init  ( & stHdmiPara ) ;
MI_HDMI_Open ( 0 ) ;
  /* Get the timing supported by the sink device */
MI_HDMI_ForceGetEdid  ( 0 , & stEdid ) ;
… _
  /* Configure according to the timing supported by the sink */
MI_MPI_HDMI_GetAttr  ( 0 , & stAttr ) ;
stAttr .  bEnableHdmi = MI_TRUE ; _
stAttr .  bEnableVideo = MI_TRUE ; _
stAttr .  eVideoFmt = enVideoFmt ; _
stAttr .  eVidOutMode = E_MI_HDMI_VIDEO_MODE_YCBCR444 ; _
stAttr .  eDeepColorMode = E_MI_HDMI_DEEP_COLOR_24BIT ; _
stAttr .  bEnableAudio = MI_FALSE ; _
stAttr .  bIsMultiChannel = MI_FALSE ; _
stAttr .  eBitDepth = E_MI_HDMI_BIT_DEPTH_16 ; _
MI_HDMI_SetAttr  ( 0 , & stAttr ) ;
MI_HDMI_Start ( 0 ) ;  _
  /* stop HDMI */
```

```
28.MI_HDMI_Stop ( 0 ) ;  _

29.MI_HDMI_Close ( 0 ) ;  _

30. MI_HDMI_DeInit () ;
```

- related topic

  MI_HDMI_DeInit [#23-mi_hdmi_deinit]

2.3. MI_HDMI_DeInit

- Function

  to initialize HDMI.

- grammar

```
MI_S32 MI_HDMI_DeInit();
```

- return value
  - 0 to initialize successfully.
  - If it is not 0, deinitialization fails. For details, refer to the error code [#jump] .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Example

  See the example of MI_HDMI_Init [#22-mi_hdmi_init] .

- related topic

  MI_HDMI_Init [#22-mi_hdmi_init]

2.4. MI_HDMI_Open

- Function

  Turn on HDMI.

- grammar

```
MI_S32 MI_HDMI_Open(MI_HDMI_DeviceId_e eHdmi);
```

- formal parameter

  Table 2-3

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter

- return value
  - 0 The HDMI device is turned on successfully.
  - Failed to open the device if it is not 0, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice

  MI_HDMI_Init [#22-mi_hdmi_init] must be called before opening HDMI [#22-mi_hdmi_init]

  Repeatedly opening HDMI returns success.

- Example

  See the example of MI_HDMI_Init [#22-mi_hdmi_init] .

- related topic

  MI_HDMI_Close [#25-mi_hdmi_close]

2.5. MI_HDMI_Close

- Function

  Turn off HDMI.

- grammar

```
MI_S32 MI_HDMI_Close(MI_HDMI_DeviceId_e eHdmi);
```

- formal parameter

  Table 2-4

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter

- return value
  - 0 Destruction of HDMI device succeeded.
  - Failed to destroy the HDMI device if it is not 0, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice

  Repeated closes return success.

- Example

  See the example of MI_HDMI_Init [#22-mi_hdmi_init] .

- related topic

  MI_HDMI_Open [#24-mi_hdmi_open]

2.6. MI_HDMI_SetAttr

- Function

  Set HDMI properties.

- grammar

```
MI_S32 MI_HDMI_SetAttr(MI_HDMI_DeviceId_e
eHdmi,MI_HDMI_Attr_t*pstAttr);
```

- formal parameter

  Table 2-5

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
pstAttr	HDMI attribute structure pointer.	enter

- return value
  - 0 Setting HDMI device properties succeeded.
  - Failed to set HDMI device properties other than 0, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - MI_HDMI_Open [#24-mi_hdmi_open] needs to be called before setting HDMI properties .
  - User needs to set HDMI properties before calling MI_HDMI_Start . [#28-mi_hdmi_start]
  - To set HDMI properties after calling MI_HDMI_Start [#28-mi_hdmi_start] , you need to call MI_HDMI_Stop [#29-mi_hdmi_stop] before setting new

HDMI properties.

- When transmitting audio, you need to ensure that video data is transmitted, and the audio is clipped into the video for transmission

- Example

```
1. MI_HDMI_InitPara_t stHdmiPara ; _
2. MI_HDMI_Attr_t stAttr ; _
3. MI_HDMI_TimingType_e enVideoFmt ; _
4. /   * Start DISP */
5 .   /* Set HDMI properties after HDMI start */
6. MI_HDMI_Stop   ( 0 , MI_TRUE ) ;
7. MI_HDMI_GetAttr   ( 0 , & stAttr ) ;
8 .   /* User changes HDMI properties such as timing */
9. stAttr .   eVideoFmt = E_MI_HDMI_VIDEO_FMT_720P_60 ; _
10. MI_HDMI_SetAttr  ( 0 , & stAttr ) ;
11. MI_HDMI_Start  ( 0 , MI_FALSE ) ;
```

- related topic

  MI_HDMI_GetAttr [#27-mi_hdmi_getattr]

2.7. MI_HDMI_GetAttr

- Function

  Get HDMI properties.

- grammar

```
MI_S32 MI_HDMI_GetAttr(MI_HDMI_DeviceId_e eHdmi, MI_HDMI_Attr_t
*pstAttr);
```

- grammar

  Table 2-6

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
pstAttr	HDMI attribute structure pointer.	output

- return value
  - 0 Getting HDMI device properties succeeded.
  - Failed to obtain HDMI device properties if not 0. For details, refer to Error Codes [#jump] .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Example

  See MI_HDMI_Init [#22-mi_hdmi_init] example.
- related topic

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr]

2.8. MI_HDMI_Start

- Function

  Start HDMI.
- grammar

```
MI_S32 MI_HDMI_Start(MI_HDMI_DeviceId_e eHdmi);
```

- formal parameter

Table 2-7

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter

- return value
  - 0 Start HDMI work successfully.
  - Non-0 HDMI device failed to start, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - MI_HDMI_Open [#24-mi_hdmi_open] must be called before starting HDMI [#24-mi_hdmi_open]
- Example

  See MI_HDMI_Init [#22-mi_hdmi_init] example.
- related topic

  MI_HDMI_Stop [#29-mi_hdmi_stop]

2.9. MI_HDMI_Stop

- Function

  Stop the HDMI device from working.
- grammar

```
MI_S32 MI_HDMI_Stop(MI_HDMI_DeviceId_e eHdmi);
```

- formal parameter

Table 2-8

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter

- return value
  - 0 stops the HDMI device from working.
  - Non-0 Failed to stop the HDMI device, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- related topic

  MI_HDMI_Start [#28-mi_hdmi_start]

2.10. MI_HDMI_GetSinkInfo

- Function

  Get the capability level information of HDMI sink side.
- grammar

```
MI_S32 MI_HDMI_GetSinkInfo(MI_HDMI_DeviceId_e eHdmi,
MI_HDMI_SinkInfo_t *pstSinkInfo);
```

- formal parameter

  Table 2-9

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
pstSinkCap	HDMI Sink side capability level structure pointer	output

- return value
  - 0 Obtaining the HDMI sink capability is successful.
  - Failed to obtain sink information if it is not 0. For details, refer to the error code [#jump] .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - HDMI must be turned on before calling. In order to ensure that the HDMI open is completed, it is recommended to wait about 500ms after calling the MI_HDMI_Open [#24-mi_hdmi_open] interface.
  - Should be called after HDMI is up and the cable is plugged in.
  - It is generally used in inserting event handlers or after forcing the EDID to be obtained.
- Example

```
MI_HDMI_Init_Para_t stHdmiPara ; _
MI_HDMI_Attr_t stAttr ; _
MI_HDMI_TimingType_e enVideoFmt ; _
MI_HDMI_SinkInfo_t stSinkInfo ; _
/   * Start DISP */
   /* Enable HDMI */
   /* Get HDMI sink capability level */
```

```
8. MI_HDMI_GetSinkInfo  ( 0 , & stSinkInfo ) ;
```

2.11. MI_HDMI_SetAvMute

- Function

  Set HDMI audio and video mute function.

- grammar

  MI_S32 MI_HDMI_SetAvMute(MI_HDMI_DeviceId_e eHdmi, MI_BOOL bAvMute);

- formal parameter

  Table 2-10

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
bAvMute	Whether to set HDMI audio and video mute. Value range: MI_TRUE: Mute audio and video. MI_FALSE: Turn off audio and video mute.	enter

- return value
  - 0 success.
  - Non-zero failure, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice

  Need to be used after calling MI_HDMI_Start [#28-mi_hdmi_start] .

- Example

```
1.MI_HDMI_Start ( 0 ) ;   _

2. MI_HDMI_SetMute   ( 0 , MI_TRUE ) ;
```

2.12. MI_HDMI_ForceGetEdid

- Function

  Get the EDID information of HDMI.

- grammar

```
MI_S32 MI_HDMI_ForceGetEdid(MI_HDMI_DeviceId_e
eHdmi,MI_HDMI_Edid_t *pstEdidData);
```

- formal parameter

  Table 2-11

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
pstEdidData	EDID information for HDMI.	output

- return value
  - 0 success.
  - If it is not 0, it fails. For details, refer to the error code [#jump] .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - HDMI must be turned on before calling.

- Example

```
MI_HDMI_InitPara_t stHdmiPara ; _
MI_HDMI_Attr_t stAttr ; _
MI_HDMI_Edid_t stEdidData ; _
/   * Start DISP */
   /* Start HDMI */
   /* Get HDMI EDID information */
MI_HDMI_ForceGetEdid   ( 0 , & stEdidData ) ;
```

2.13. MI_HDMI_SetDeepColor

- Function

  Set the Deep Color mode.

- grammar

```
MI_S32 MI_HDMI_SetDeepColor(MI_HDMI_DeviceId_e
eHdmi,MI_HDMI_DeepColor_e eDeepColor);
```

- formal parameter

  Table 2-12

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	
enter		
eDeepColor	Deep Color mode for HDMI.	enter

- return value

-0 for success.

- Non-zero failure, please refer to the error code [#jump] for details .

- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - This interface is a high-level interface and generally does not need to be called.
  - HDMI must be turned on before calling.
  - This interface can be used to set some Deep Color modes not supported by MI_HDMI_SetAttr . [#26-mi_hdmi_setattr]
  - This interface has not been implemented yet.
- Example

```
MI_HDMI_InitPara_t stHdmiPara ; _
MI_HDMI_Attr_t stAttr ; _
    /* Start DISP*/
    /* Enable HDMI */
    /* Set Deep color mode */
MI_HDMI_SetDeepColor   ( 0 , E_MI_HDMI_DEEP_COLOR_24BIT  )
;
```

2.14. MI_HDMI_GetDeepColor

- Function

  Get the Deep Color mode.

- grammar

```
MI_S32 MI_HDMI_GetDeepColor(MI_HDMI_DeviceId_e eHdmi,
MI_HDMI_DeepColor_e *peDeepColor);
```

- formal parameter

  Table 2-13

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
peDeepColor	HDMI's Deep Color mode pointer.	enter

- return value
  - 0 success.
  - Non-zero initialization failed, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - This interface is a high-level interface and generally does not need to be called.
  - HDMI must be turned on before calling.
  - This interface has not been implemented yet.
- related topic

  MI_HDMI_SetDeepColor [#213-mi_hdmi_setdeepcolor]

2.15. MI_HDMI_SetInfoFrame

- Function

  Set HDMI info frame.

- grammar

```
MI_S32 MI_HDMI_SetInfoFrame(MI_HDMI_DeviceId_e
eHdmi,MI_HDMI_InfoFrame_t *pstInfoFrame);
```

- formal parameter

  Table 2-14

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
pstInfoFrame	Image infoframe structure pointer.	enter

- return value
  - 0 success.
  - Non-zero failure, please refer to the error code [#jump] for details .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - This interface is generally used to configure information frame properties in user mode callback functions. In other cases, it is generally not necessary to call this interface.
  - HDMI must be turned on before calling.
  - This interface only supports settings `AVI` , `SPD` and `AUDIO` information frames, other types of information frame settings are invalid.
- Example

```
1. MI_HDMI_DeviceId_e eHdmi = E_MI_HDMI_ID_0 ; _

2. MI_HDMI_InfoFrame_t stAviInfoFrame = {
E_MI_HDMI_INFOFRAME_TYPE_AVI } ; _
```

```
3. MI_HDMI_InfoFrame_t stAudInfoFrame = {
E_MI_HDMI_INFOFRAME_TYPE_AUDIO } ; _

4. MI_HDMI_InfoFrame_t stSpdInfoFrame = {
E_MI_HDMI_INFOFRAME_TYPE_SPD } ; _

5. /   * AviInfoFrame */

6. stAviInfoFrame .   unInforUnit . stAviInfoFrame .
bEnableAfdOverWrite = 0 ; _

7. …   _

8. /   * AudioInfoFrame */

9. stAudInfoFrame .   unInforUnit . stAudInfoFrame .
u32ChannelCount = 2 ; _

10. … _

11. /  * SpdInfoFrame */

12. stSpdInfoFrame .  unInforUnit . stSpdInfoFrame .
au8VendorName [ 0 ] = 0 ; _

13. …  _

14. MI_HDMI_SetInfoFrame  ( eHdmi , & stAviInfoFrame ) ;
```

- related topic

  MI_HDMI_GetInfoFrame [#216-mi_hdmi_getinfoframe]

2.16. MI_HDMI_GetInfoFrame

- Function

  Get HDMI infoframe.

- grammar

```
MI_RESULT MI_HDMI_GetInfoFrame(MI_HDMI_DeviceId_e
eHdmi,MI_HDMI_InfoFrameType_e
eInfoFrameType,MI_HDMI_InfoFrame_t *pstInfoFrame);
```

- formal parameter

  Table 2-15

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
eInfoFrameType	Information frame type.	enter
pstInfoFrame	Image infoframe structure pointer.	enter

- return value
  - 0 Initialization succeeded.
  - If it is not 0, initialization fails. For details, refer to the error code [#jump] .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice

  This interface is generally used to obtain information frame attributes in user mode callback functions. In other cases, it is generally not necessary to call this interface.

2.17. MI_HDMI_SetAnalogDrvCurrent

- Function

  Set the drive current of the HDMI channel

- grammar

```
MI_RESULT MI_HDMI_SetAnalogDrvCurrent(MI_HDMI_DeviceId_e
eHdmi,MI_HDMI_AnalogDrvCurrent_t *pstAnalogDrvCurrent);
```

- formal parameter

  Table 2-16

parameter name	describe	input Output
eHdmi	HDMI port number. Value range: 0	enter
pstAnalogDrvCurrent	Structure type of HDMI output current	enter

- return value
  - 0 Initialization succeeded.
  - If it is not 0, initialization fails. For details, refer to the error code [#jump] .
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - This interface is an advanced interface, providing users to adjust the output current of the HDMI channel.
  - HDMI must be turned on before calling.
- Example

```
1. MI_HDMI_AnalogDrvCurrent_t  CurInfo  =  {  } ;
2. memset ( &  CurInfo , 0xFF , sizeof ( CurInfo ) ) ;
3. MI_HDMI_SetAnalogDrvCurrent ( g_HdmiAttr .  eHdmi , &
CurInfo ) ;
```

2.18. MI_HDMI_InitDev

- Function

initialize hdmi device

- grammar

```
MI_S32 MI_HDMI_InitDev(MI_HDMI_InitParam_t *pstInitParam);
```

- formal parameter

Table 2-17

parameter name	describe	input Output
pstInitParam	Initialize the parameter structure pointer, which can be configured as NULL	enter

- return value
  - 0 Initialization succeeded.
  - If it is not 0, the initialization fails. For details, refer to the error code.
- rely
  - Header files: mi_hdmi.h, mi_hdmi_datatype.h
  - Library file: libmi_hdmi.a
- Notice
  - This interface is recommended for Version 2.06 and above to replace the original MI_HDMI_Init interface.

2.19. MI_HDMI_DeInitDev

- Function

Deinitialize HDMI device

- grammar

```
MI_S32 MI_HDMI_DeInitDev(void);
```

- return value
    - 0 Initialization succeeded.
    - If it is not 0, the initialization fails. For details, refer to the error code.
- rely
    - Header files: mi_hdmi.h, mi_hdmi_datatype.h
    - Library file: libmi_hdmi.a
- Notice

  This interface is recommended for Version 2.06 and above to replace the original MI_HDMI_DeInit [#23-mi_hdmi_deinit] interface.

---

3. HDMI data type

3.1. Data Type Definition

Table 3-1

type of data	definition
MI_HDMI_DeviceId_e [#32-mi_hdmi_deviceid_e]	Define the HDMI interface number
MI_HDMI_InitPara_t [#33-mi_hdmi_initpara_t]	Define the HDMI initialization properties structure
MI_HDMI_EventCallBack [#34-mi_hdmi_eventcallback]	Define HDMI callback function
MI_HDMI_EventType_e [#35-mi_hdmi_eventtype_e]	Define HDMI event notification type
MI_HDMI_Attr_t [#36-mi_hdmi_attr_t]	Define HDMI Properties Structure
MI_HDMI_TimingType_e [#37-mi_hdmi_timingtype_e]	Define HDMI video format type
MI_HDMI_OutputMode_e [#38-	Define HDMI color space type

type of data definition

MI_HDMI_OutputMode_e [#38-mi_hdmi_outputmode_e]	Define HDMI color space type
MI_HDMI_ColorType_e [#39-mi_hdmi_colortype_e]	Define HDMI output color type
MI_HDMI_DeepColor_e [#310-mi_hdmi_deepcolor_e]	Define HDMI dark mode type
MI_HDMI_SampleRate_e [#311-mi_hdmi_samplerate_e]	Defines the HDMI audio output sample rate type
MI_HDMI_BitDepth_e [#312-mi_hdmi_bitdepth_e]	Define HDMI audio output sampling bit width type
MI_HDMI_SinkInfo_t [#313-mi_hdmi_sinkinfo_t]	Define the HDMI Sink side capability level structure
MI_HDMI_Edid_t [#314-MI_HDMI_Edid_t]	Define HDMI EDID information structure
MI_HDMI_InfoFrameType_e [#315-mi_hdmi_infoframetype_e]	Define HDMI infoframe type
MI_HDMI_AviInfoFrameVer_t [#316-mi_hdmi_aviinfoframever_t]	Define AVI infoframe structure
MI_HDMI_AudInfoFrameVer_t [#317-mi_hdmi_audinfoframever_t]	Define the AUDIO information frame structure
MI_HDMI_SpdInfoFrame_t [#318-mi_hdmi_spdinfoframe_t]	Defining the SPD Information Frame Structure
MI_HDMI_InfoFrame_Unit_u [#319-mi_hdmi_infoframe_unit_u]	Defining the HDMI InfoFrame Consortium
MI_HDMI_AnalogDrvCurrent_t [#320-mi_hdmi_analogdrvcurrent_t]	Define HDMI channel drive current structure

Note: This section has covered all important data types, some data types not listed, please refer to mi_ hdmi_datatype.h

3.2. MI_HDMI_DeviceId_e

- illustrate

  Define the HDMI port number.

- definition

```
typedef  enum

{

    E_MI_HDMI_ID_0  =  0 ,

    E_MI_HDMI_ID_BUTT

}  MI_HDMI_DeviceId_e ;
```

- member

  Table 3-2

member name	describe
E_MI_HDMI_ID_0	HDMI port 0

3.3. MI_HDMI_InitPara_t

- illustrate

  Defines the HDMI initialization properties structure.

- definition

```
typedef  struct  MI_HDMI_InitPara_s

{

    MI_HDMI_EventCallBack  pfnHdmiEventCallback ;

    void  * pCallBackArgs ;
```

```
} MI_HDMI_InitPara_t ;
```

- member

Table 3-3

member name	describe
pfnHdmiEventCallback	event handler callback function
pCallBackArgs	Callback function parameter private data

- Related data types and interfaces

MI_HDMI_Init [#22-mi_hdmi_init]

3.4. MI_HDMI_EventCallBack

- illustrate

Define the HDMI callback function.

- grammar

```
typedef MI_RESULT (*MI_HDMI_EventCallBack) (MI_HDMI_DeviceId_e
eHdmi, MI_HDMI_EventType_e event, void entParam, void
*pUsrParam);
```

- member

Table 3-4

member name	describe
MI_HDMI_EvenType_e	HDMI event type
pEventParam pUsrParam	Event private data

3.5. MI_HDMI_EventType_e

- illustrate

  Defines the HDMI event notification type.

- definition

```
typedef  enum

{

    E_MI_HDMI_EVENT_HOTPLUG  =  0 ,

    E_MI_HDMI_EVENT_NO_PLUG ,

    E_MI_HDMI_EVENT_MAX

}  MI_HDMI_EventType_e ;
```

- member

  Table 3-5

member name	describe
E_MI_HDMI_EVENT_HOTPLUG	HDMI Cable connection event
E_MI_HDMI_EVENT_NO_PLUG	HDMI Cable disconnect event

- Related data types and interfaces

  MI_HDMI_Init [#22-mi_hdmi_init]

3.6. MI_HDMI_Attr_t

- illustrate

  Defines the HDMI attribute structure.

- definition

```
typedef  struct  MI_HDMI_Attr_s

{

    MI_HDMI_VideoAttr_t  stVideoAttr ;

    MI_HDMI_AudioAttr_t  stAudioAttr ;

    MI_HDMI_EnInfoFrame_t  stEnInfoFrame ;

}  MI_HDMI_Attr_t ;

typedef  struct  MI_HDMI_VideoAttr_s

{

    MI_BOOL  bEnableVideo ;

    MI_HDMI_TimingType_eTimingType  ; _

    MI_HDMI_OutputMode_e  eOutputMode ;

    MI_HDMI_ColorType_e  eColorType ;

    MI_HDMI_ColorType_e  eInColorType ;

    MI_HDMI_DeepColor_e  eDeepColorMode ;

}  MI_HDMI_VideoAttr_t ;

typedef  struct  MI_HDMI_AudioAttr_s

{

    MI_BOOL  bEnableAudio ;

    MI_BOOL  bIsMultiChannel ;

    MI_HDMI_SampleRate_e  eSampleRate ;

    MI_HDMI_BitDepth_e  eBitDepth ;

}  MI_HDMI_VideoAttr_t ;

    typedef  struct  MI_HDMI_EnInfoFrame_s

{
```

```
    MI_BOOL  bEnableAviInfoFrame ;

    MI_BOOL  bEnableAudInfoFrame ;

    MI_BOOL  bEnableSpdInfoFrame ;

    MI_BOOL  bEnableMpegInfoFrame ;

} MI_HDMI_EnInfoFrame_t ;
```

- member

  Table 3-6

member name	describe
bEnableVideo	Whether to output video
eTimingType	Video format, this parameter needs to be consistent with the format of the video output configuration
eOutputMode	HDMI output video mode
eColorType	HDMI output color type
eInColorType	HDMI input color type
eDeepColorMode	DeepColor output mode. Default is MI_HDMI_DEEP_COLOR_24BIT Not supported. MI_HDMI_DEEP_COLOR_30BIT and MI_HDMI_DEEP_COLOR_36BIT. MI_HDMI_DEEP_COLOR_48BIT
bEnableAudio	Whether to enable audio
bIsMultiChannel	Multi-channel number 0: 2 channels; 1: 8 channels. /not currently supported
enSampleRate	Audio sampling rate, this parameter needs to be consistent with the AO configuration
enBitDepth	Audio bit width, the default is 16, this parameter needs to be consistent with the AO configuration
bEnableAviInfoFrame	Whether to enable AVI InfoFrame is recommended to enable
bEnableAudInfoFrame	Whether to enable AUDIO InfoFrame, it is recommended to enable
bEnableSpdInfoFrame	Whether to enable SPD InfoFrame

- Precautions

  - User can set HDMI properties according to suggested values

- Related data types and interfaces

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr]

3.7. MI_HDMI_TimingType_e

- illustrate

  Defines the HDMI video format type.

- definition

```
typedef  enum

{

    E_MI_HDMI_TIMING_480_60I  =  0 ,

    E_MI_HDMI_TIMING_480_60P  =  1 ,

    E_MI_HDMI_TIMING_576_50I  =  2 ,

    E_MI_HDMI_TIMING_576_50P  =  3 ,

    E_MI_HDMI_TIMING_720_50P  =  4 ,

    E_MI_HDMI_TIMING_720_60P  =  5 ,

    E_MI_HDMI_TIMING_1080_50I  =  6 ,

    E_MI_HDMI_TIMING_1080_50P  =  7 ,

    E_MI_HDMI_TIMING_1080_60I  =  8 ,

    E_MI_HDMI_TIMING_1080_60P  =  9 ,

    E_MI_HDMI_TIMING_1080_30P  =  10 ,

    E_MI_HDMI_TIMING_1080_25P  =  11 ,

    E_MI_HDMI_TIMING_1080_24P  =  12 ,

    E_MI_HDMI_TIMING_4K2K_30P  =  13 ,
```

```
    E_MI_HDMI_TIMING_1440_50P  =  14 ,

    E_MI_HDMI_TIMING_1440_60P  =  15 ,

    E_MI_HDMI_TIMING_1440_24P  =  16 ,

    E_MI_HDMI_TIMING_1440_30P  =  17 ,

    E_MI_HDMI_TIMING_1470_50P  =  18 ,

    E_MI_HDMI_TIMING_1470_60P  =  19 ,

    E_MI_HDMI_TIMING_1470_24P  =  20 ,

    E_MI_HDMI_TIMING_1470_30P  =  21 ,

    E_MI_HDMI_TIMING_1920x2205_24P  =  22 ,

    E_MI_HDMI_TIMING_1920x2205_30P  =  23 ,

    E_MI_HDMI_TIMING_4K2K_25P  =  24 ,

    E_MI_HDMI_TIMING_4K1K_60P  =  25 ,

    E_MI_HDMI_TIMING_4K2K_60P  =  26 ,

    E_MI_HDMI_TIMING_4K2K_24P  =  27 ,

    E_MI_HDMI_TIMING_4K2K_50P  =  28 ,

    E_MI_HDMI_TIMING_2205_24P  =  29 ,

    E_MI_HDMI_TIMING_4K1K_120P  =  30 ,

    E_MI_HDMI_TIMING_4096x2160_24P  =  31 ,

    E_MI_HDMI_TIMING_4096x2160_25P  =  32 ,

    E_MI_HDMI_TIMING_4096x2160_30P  =  33 ,

    E_MI_HDMI_TIMING_4096x2160_50P  =  34 ,

    E_MI_HDMI_TIMING_4096x2160_60P  =  35 ,

    E_MI_HDMI_TIMING_1024x768_60P  =  36 ,

    E_MI_HDMI_TIMING_1280x1024_60P  =  37 ,

    E_MI_HDMI_TIMING_1440x900_60P  =  38 ,
```

```
    E_MI_HDMI_TIMING_1600x1200_60P  =  39 ,

    E_MI_HDMI_TIMING_1280x800_60P  =  40 ,

    E_MI_HDMI_TIMING_1366x768_60P  =  41 ,

    E_MI_HDMI_TIMING_1680x1050_60P  =  42 ,

    E_MI_HDMI_TIMING_1920x1080_5994P  =  43 ,

    E_MI_HDMI_TIMING_1920x1080_2997P  =  44 ,

    E_MI_HDMI_TIMING_1280x720_5994P   =  45 ,

    E_MI_HDMI_TIMING_1280x720_2997P   =  46 ,

    E_MI_HDMI_TIMING_MAX ,

}  MI_HDMI_TimingType_e ;
```

- Precautions
  - It is necessary to set the corresponding HDMI format according to the format of the video output.
  - When the AP is setting the timing, it needs to match the timing of the DISP module. The timing of the HDMI must be consistent with the timing of the DISP module.
- Related data types and interfaces

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr]

3.8. MI_HDMI_OutputMode_e

- illustrate

  Define HDMI output mode type

- definition

```
typedef  enum

{
```

```
    E_MI_HDMI_OUTPUT_MODE_HDMI  =  0 ,

    E_MI_HDMI_OUTPUT_MODE_HDMI_HDCP ,

    E_MI_HDMI_OUTPUT_MODE_DVI ,

    E_MI_HDMI_OUTPUT_MODE_DVI_HDCP ,

    E_MI_HDMI_OUTPUT_MODE_MAX ,

}  MI_HDMI_OutputMode_e ;
```

- member

  Table 3-7

member name	describe
E_MI_HDMI_OUTPUT_MODE_HDMI	HDMI output mode
E_MI_HDMI_OUTPUT_MODE_HDMI_HDCP	HDMI + DHCP output mode
E_MI_HDMI_OUTPUT_MODE_DVI	DVI output mode
E_MI_HDMI_OUTPUT_MODE_DVI_HDCP	DVI + DHCP output mode

- Related data types and interfaces

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr]

3.9. MI_HDMI_ColorType_e

- illustrate

  Defines the HDMI color space type.

- definition

```
typedef  enum

{
```

```
    E_MI_HDMI_COLOR_TYPE_RGB444  =  0 ,
    E_MI_HDMI_COLOR_TYPE_YCBCR422 ,
    E_MI_HDMI_COLOR_TYPE_YCBCR444 ,
    E_MI_HDMI_COLOR_TYPE_YCBCR420 ,
    E_MI_HDMI_COLOR_TYPE_MAX
}  MI_HDMI_ColorType_e ;
```

- member

  Table 3-8

member name	describe
E_MI_HDMI_COLOR_TYPE_RGB444	RGB444 output mode
E_MI_HDMI_COLOR_TYPE_YCBCR422	YCBCR422 output mode
E_MI_HDMI_COLOR_TYPE_YCBCR444	YCBCR444 output mode
E_MI_HDMI_COLOR_TYPE_YCBCR420	YCBCR420 output mode

- Related data types and interfaces

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr]

3.10. MI_HDMI_DeepColor_e

- illustrate

  Defines the HDMI dark mode type.

- definition

```
typedef  enum
{
```

```
    E_MI_HDMI_DEEP_COLOR_24BIT  =  0x00 , _
    E_MI_HDMI_DEEP_COLOR_30BIT ,
    E_MI_HDMI_DEEP_COLOR_36BIT ,
    E_MI_HDMI_DEEP_COLOR_48BIT ,
    E_MI_HDMI_DEEP_COLOR_MAX ,
}  MI_HDMI_DeepColor_e ;
```

- member

  Table 3-9

member name	describe
E_MI_HDMI_DEEP_COLOR_24BIT	HDMI Deep Color 24bit mode
E_MI_HDMI_DEEP_COLOR_30BIT	HDMI Deep Color 30bit mode
E_MI_HDMI_DEEP_COLOR_36BIT	HDMI Deep Color 36bit mode
E_MI_HDMI_DEEP_COLOR_48BIT	HDMI Deep Color 48bit mode

- Related data types and interfaces

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr]

3.11. MI_HDMI_SampleRate_e

- illustrate

  Defines the HDMI audio output sample rate type.

- definition

```
typedef  enum
{
```

```
    E_MI_HDMI_AUDIO_SAMPLERATE_UNKNOWN  =  0 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_32K  =  1 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_44K  =  2 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_48K  =  3 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_88K  =  4 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_96K  =  5 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_176K  =  6 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_192K  =  7 ,

    E_MI_HDMI_AUDIO_SAMPLERATE_MAX ,

}  MI_HDMI_SampleRate_e ;
```

- Precautions
  - When the AP is setting the sampling rate, it needs to match the sampling rate of the Audio module. The sampling rate of HDMI must be consistent with the sampling rate of the Audio module.
- Related data types and interfaces

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr]

3.12. MI_HDMI_BitDepth_e

- illustrate

  Defines the HDMI audio output sampling bit width type.

- definition

```
typedef  enum

{

    E_MI_HDMI_BIT_DEPTH_UNKNOWN  = 0 ,
```

```
    E_MI_HDMI_BIT_DEPTH_8  =  8 ,

    E_MI_HDMI_BIT_DEPTH_16  =  16 ,

    E_MI_HDMI_BIT_DEPTH_18  =  18 ,

    E_MI_HDMI_BIT_DEPTH_20  =  20 ,

    E_MI_HDMI_BIT_DEPTH_24  =  24 ,

    E_MI_HDMI_BIT_DEPTH_32  =  32 ,

    E_MI_HDMI_BIT_DEPTH_BUTT

} MI_HDMI_BitDepth_e ;
```

- Related data types and interfaces

  MI_HDMI_SetAttr [#26-mi_hdmi_setattr] v

3.13. MI_HDMI_SinkInfo_t

- illustrate

  Define the HDMI Sink side capability level structure.

- definition

```
typedef struct MI_HDMI_Sink_Info_s

{

    MI_BOOL bConnected ;

    MI_BOOL bSupportHdmi ;

    MI_HDMI_TimingType_e eNativeTimingType ;

    MI_BOOL abVideoFmtSupported [ E_MI_HDMI_TIMING_MAX ] ;

    MI_BOOL bSupportYCbCr444 ;

    MI_BOOL bSupportYCbCr422 ;

    MI_BOOL bSupportYCbCr ;
```

```
    MI_BOOL bSupportxvYcc601 ;

    MI_BOOL bSupportxvYcc709 ;

    MI_U8 u8MdBit ;

    MI_BOOL abAudioFmtSupported [ MI_HDMI_MAX_ACAP_CNT ] ;

    MI_U32 au32AudioSampleRateSupported [
MI_HDMI_MAX_AUDIO_SAMPLE_RATE_CNT ] ;

    MI_U32 u32MaxPcmChannels ;

    MI_U8 u8Speaker ;

    MI_U8 au8IdManufactureName [ 4 ] ;

    MI_U32 u32IdProductCode ;

    MI_U32 u32IdSerialNumber ;

    MI_U32 u32WeekOfManufacture ;

    MI_U32 u32YearOfManufacture ;

    MI_U8 u8Version ;

    MI_U8 u8Revision ;

    MI_U8 u8EdidExternBlockNum ;

    MI_U8 au8IeeRegId [ 3 ] ; // IEEE registration identifier


    MI_U8 u8PhyAddr_A ;

    MI_U8 u8PhyAddr_B ;

    MI_U8 u8PhyAddr_C ;

    MI_U8 u8PhyAddr_D ;

    MI_BOOL bSupportDviDual ;

    MI_BOOL bSupportDeepColorYcbcr444 ;

    MI_BOOL bSupportDeepColor30Bit ;
```

```
    MI_BOOL bSupportDeepColor36Bit ;

    MI_BOOL bSupportDeepColor48Bit ;

    MI_BOOL bSupportAi ;

    MI_U32 u8MaxTmdsClock ;

    MI_BOOL bILatencyFieldsPresent ;

    MI_BOOL bLatencyFieldsPresent ;

    MI_BOOL bHdmiVideoPresent ;

    MI_U8 u8VideoLatency ;

    MI_U8 u8AudioLatency ;

    MI_U8 u8InterlacedVideoLatency ;

    MI_U8 u8InterlacedAudioLatency ;

} MI_HDMI_SinkInfo_t ;
```

- member

  Table 3-10

member name	describe
bConnected	Is the device connected
bSupportHdmi	Whether the device supports HDMI, if not, it is a DVI device
eNativeVideoFormat	Display device physical resolution.
abVideoFmtSupported	Video Capability Set. MI_TRUE: This display format is supported; MI_FALSE: Not supported.
bSupportYCbCr	Whether to support YCBCR display. MI_TRUE: Support YCBCR display; MI_FALSE: Only support RGB display.
bSupportxvYCC601	Whether to support the xvYCC601 color format

member name describe

member name	describe
bSupportxvYCC601	Whether to support the xvYCC601 color format
bSupportxvYCC709	Whether to support xvYCC709 color format
u8MDBit	Transmission profiles supported by xvYCC601: 1:P0,2:P1,4:P2
bAudioFmtSupported	For audio capability set, please refer to EIA-CEA-861-D Table 37 MI_TRUE: this display format is supported; MI_FALSE: not supported.
u32AudioSampleRateSupported	Audio sample rate capability set, 0 is an illegal value, others are supported audio sample rates.
u32MaxPcmChannels	The maximum number of PCM channels for audio.
u8Speaker	For speaker location, please refer to the definition of SpeakerDATABlock in EIA-CEA-861-D.
u8IDManufactureName	Device manufacturer identification.
u32IDProductCode	Device ID
u32IDSerialNumber	devise serial number
u32WeekOfManufacture	Equipment production date (week)
u32YearOfManufacture	Equipment production date (year)
u8Version	Device version number
u8Revision	Device subversion number
au8EDIDExternBlockNum	EDID Expansion Block Number
au8IEERegId	24-bit IEEE Registration Identifier (0x000C03).
u8PhyAddr_A	CEC Physical Address A
u8PhyAddr_B	CEC Physical Address B
u8PhyAddr_C	CEC physical address C

u8PhyAddr_C	CEC physical address C
u8PhyAddr_D	CEC physical address D
bSupportDVIDual	Whether to support DVI dual-link operation.
bSupportDeepColorYCBCR444	Whether to support YCBCR 4:4:4 Deep Color mode.
bSupportDeepColor36Bit	Whether to support Deep Color 36bit mode
bSupportDeepColor48Bit	Whether to support Deep Color 48bit mode
bSupportAI	Whether to support Supports_AI mode
u8MaxTMDSClock	Maximum TMDS Clock
bLatencyFieldsPresent	Delay flag bit
bHDMIVideoPresent	Whether the Video_Latency and Audio_Latency fields exist.
u8VideoLatency	video delay
u8AudioLatency	audio delay
u8InterlacedVideoLatency	Video delay in interlaced video mode
u8InterlacedAudioLatency	Audio delay in interlaced video mode

- Related data types and interfaces

  MI_HDMI_GetSinkInfo [#210-mi_hdmi_getsinkinfo]

3.14. MI_HDMI_Edid_t

- illustrate

  Define HDMI EDID information structure

- definition

```
typedef  struct  MI_HDMI_Edid_s
{
    MI_BOOL  bEdidValid ;
    MI_U32  u32Editlength ;
    MI_U8  au8Edit [ 512 ];
}  MI_HDMI_Edid_t ;
```

- member

  Table 3-11

member name	describe
bEdidValid	Is the EDID information valid?
u32Editlength	EDID information data length, generally 256 Byte
au8Edid	au8Edid

- Related data types and interfaces

  MI_HDMI_ForceGetEdid [#212-mi_hdmi_forcegetedid]

3.15. MI_HDMI_InfoFrameType_e

- illustrate

  Defines the HDMI infoframe type.

- definition

```
typedef  enum
{
    E_MI_INFOFRAME_TYPE_AVI  =  0 ,
```

```
    E_MI_INFOFRAME_TYPE_SPD ,

    E_MI_INFOFRAME_TYPE_AUDIO ,

    E_MI_INFOFRAME_TYPE_MAX

} MI_HDMI_InfoFrameType_e ;
```

- member

  Table 3-12

member name	describe
E_MI_INFOFRAME_TYPE_AVI	AVI infoframe type
E_MI_INFOFRAME_TYPE_SPD	SPD info frame type
E_MI_INFOFRAME_TYPE_AUDIO	AUDIO info frame type

- Related data types and interfaces

  MI_HDMI_DeInit [#23-mi_hdmi_deinit]

  MI_HDMI_SetInfoFrame [#215-mi_hdmi_setinfoframe]

3.16. MI_HDMI_AviInfoFrameVer_t

- illustrate

  Defines the AVI infoframe structure.

- definition

```
typedef struct MI_HDMIAviInforFrameVer_s

{

    MI_BOOL enableAFDoverWrite ;

    MI_U8 A0Value ;

    MI_HDMI_ColorType_e eColorType ;
```

```
    MI_HDMI_Colorimetry_e  eColorimetry ;

    MI_HDMI_ExtColorimetry_e  eExtColorimetry ;

    MI_HDMI_YccQuantRange_e  eYccQuantRange ;

    MI_HDMI_TimingType_eTimingType  ; // trans to
MS_VIDEO_TIMING in impl _

    MI_HDMI_VideoAfdRatio_e  eAfdRatio ;

    MI_HDMI_ScanInfo_e  eScanInfo ;

    MI_HDMI_AspectRatio_e  eAspectRatio ;

}  MI_HDMI_AviInfoFrameVer_t ;
```

- member

  Table 3-13

member name	describe
eColorType	color space type
eScanInfo	scan information enumeration variable
eAspectRatio	Aspect ratio enumeration variable
eExtColorimetry	Extended color space matrix
eTimingType	Video format type
eYccQuantRange	YCC color space range

- Precautions
  - The enumeration members of each member variable are detailed in the mi_hdmi_datatype.h file.
- Related data types and interfaces

  MI_HDMI_SetInfoFrame [#215-mi_hdmi_setinfoframe]

MI_HDMI_GetInfoFrame [#216-mi_hdmi_getinfoframe]

3.17. MI_HDMI_AudInfoFrameVer_t

- illustrate

  Define the AUDIO info frame structure.

- definition

```
typedef struct MI_HDMI_AudInfoFrameVer_s
{
    MI_U32 u32ChannelCount ;
    MI_HDMI_AudioCodeType_e eAudioCodeType ;
    MI_HDMI_SampleRate_e eSampleRate ;
} MI_HDMI_AudInfoFrameVer_t ;
```

- member

  Table 3-14

member name	describe
u32ChannelCount	number of channels.
eAudioCodeType	Audio data encoding type.
eSampleRate	Sampling frequency. It is recommended to set 0 0: the table is determined by the code stream header. 1: 32kHz 2: 44.1 kHz 3: 48 kHz 4: 88.2 kHz 5: 96 kHz 6: 176.4 kHz 7: 192 kHz

- Related data types and interfaces

  MI_HDMI_SetInfoFrame [#215-mi_hdmi_setinfoframe]

  MI_HDMI_GetInfoFrame [#216-mi_hdmi_getinfoframe]

3.18. MI_HDMI_SpdInfoFrame_t

- illustrate

  Define the SPD infoframe structure.

- definition

```
typedef struct MI_HDMI_SpdInfoFrame_s

{

    MI_U8 au8VendorName [ 8 ];

    MI_U8 au8ProductDescription [ 16 ];

} MI_HDMI_SpdInfoFrame_t ;
```

- member

  Table 3-15

member name	describe
au8VendorName[8]	seller name
au8ProductDescription[16]	product descriptor

- Related data types and interfaces

  MI_HDMI_SetInfoFrame [#215-mi_hdmi_setinfoframe]

  MI_HDMI_GetInfoFrame [#216-mi_hdmi_getinfoframe]

3.19. MI_HDMI_InfoFrame_Unit_u

- illustrate

  Defines the HDMI InfoFrame complex.

- definition

```
typedef  union

{

    MI_HDMI_Avi  InfoFrameVer_t  stAviInfoFrame ;

    MI_HDMI_AudInfoFrameVer_t  stAudInfoFrame ;

    MI_HDMI_SpdInfoFrame_t  stSpdInfoFrame ;

}  MI_HDMI_InfoFrameUnit_u ;
```

- member

  Table 3-16

member name	describe
stAviInfoFrame	AVI InfoFrame
stAudInfoFrame	AUD info frame
stSpdInfoFrame	SPD info frame

- Related data types and interfaces

  MI_HDMI_SetInfoFrame [#215-mi_hdmi_setinfoframe]

  MI_HDMI_GetInfoFrame [#216-mi_hdmi_getinfoframe]

3.20. MI_HDMI_AnalogDrvCurrent_t

- illustrate

  Defines the HDMI channel drive current structure.

- definition

```
typedef  struct  MI_HDMI_AnalogDrvCurrent_s

{

    MI_U8  u8DrvCurTap1Ch0 ;

    MI_U8  u8DrvCurTap1Ch1 ;

    MI_U8  u8DrvCurTap1Ch2 ;

    MI_U8  u8DrvCurTap1Ch3 ;

    MI_U8  u8DrvCurTap2Ch0 ;

    MI_U8  u8DrvCurTap2Ch1 ;

    MI_U8  u8DrvCurTap2Ch2 ;

    MI_U8  u8DrvCurTap2Ch3 ;

}  MI_HDMI_AnalogDrvCurrent_t ;
```

- member

  Table 3-17

member name	describe
u8DrvCurTap1Ch0	Tap1 Channel 0 Drive Current Configuration
u8DrvCurTap1Ch1	Tap1 Channel 1 Drive Current Configuration
u8DrvCurTap1Ch2	Tap1 Channel 2 Drive Current Configuration
u8DrvCurTap1Ch3	Tap1 Channel 3 Drive Current Configuration
u8DrvCurTap2Ch0	Tap2 Channel 0 Drive Current Configuration
u8DrvCurTap2Ch1	Tap2 Channel 1 Drive Current Configuration
u8DrvCurTap2Ch2	Tap2 Channel 2 Drive Current Configuration
u8DrvCurTap2Ch3	Tap2 Channel 3 Drive Current Configuration

- Related data types and interfaces

  MI_HDMI_SetAnalogDrvCurrent [#217-mi_hdmi_setanalogdrvcurrent]

4. Error code

Table 4-1 HDMI API error codes

error code	macro definition	describe
0xA00B2011	MI_ERR_HDMI_INVALID_PARAM	illegal parameter
0xA00B2001	MI_ERR_HDMI_INVALID_DEVID	invalid device id
0xA00B2018	MI_ERR_HDMI_DRV_FAILED	HDMI driver returns failure
0xA00B2015	MI_ERR_HDMI_NOT_INIT	HDMI not initialized
0xA00B2008	MI_ERR_HDMI_NOT_SUPPORT	Operation not supported
0xA00B21E0	MI_ERR_HDMI_UNSUPPORT_TIMING	Output timing is not supported
0xA00B21E1	MI_ERR_HDMI_UNSUPPORT_COLORTYPE	Color type not supported
0xA00B21E2	MI_ERR_HDMI_UNSUPPORT_ACODETYPE	Unsupported audio encoding type
0xA00B21E3	MI_ERR_HDMI_UNSUPPORT_AFREQ	unsupported audio frequency
0xA00B21E4	MI_ERR_HDMI_EDID_HEADER_ERR	EDID header information parsing failed
0xA00B21E5	MI_ERR_HDMI_EDID_DATA_ERR	EDID data error

5. Introduction to PROCFS

5.1. cat

- debug info

```
# cat / proc / mi_modules / mi_hdmi / mi_hdmi0
```

```
------------------------------------ HDMI0 Dev Info --------
----------------------------
InitFlag   AvMute   PowerOn
    Y        N         Y
EnableVideo   TimingType   OutputMode   ColorType
DeepColorMode
    1            9            0            2            4
EnableAudio   IsMultiChannel   BitDepth   CodeType   SampleRate
    1            0            16           0            3
```

- Analysis of debug information

  Recording the current HDMI usage and device properties can dynamically obtain this information, which is convenient for debugging and testing.

- Parameter Description

  </table

parameter		describe
device info	InitFlag	Whether the HDMI module is initialized Y: Initialized N: Not initialized
	AvMute	Whether the audio and video is in the MUTE state Y: MUTE N: Not MUTE
	PowerOn	Whether HDMI enables power management Y: Yes N: No
	EnableVideo	Whether to enable video 0: not enabled 1: enabled
	TimingType	Current HDMI output resolution Value range [0~ E_MI_HDMI_TIMING_MAX]
	OutputMode	HDMI output mode 0: HDMI mode 2: DVI mode 1 / 3 is HDCP mode, currently not supported

ColorType	Color space 0: RGB44 1: YUV422 2: YUV444 3: YUV420
DeepColorMode	Color bit depth 0: 8Bit 1: 10Bit 2: 12Bit 3: 16Bit
EnableAudio	Whether to enable audio 0: not enabled 1: enabled
IsMultiChannel	Whether multi-channel 0: No 1: Yes
BitDepth	Audio sampling bit depth 8: 8Bit 16: 16Bit 32: 32Bit
CodeType	Compression format for audio output 0: PCM 1: Non-PCM
SampleRate	Audio output sample rate 0: unknown 1: 32K 2: 44K 3: 48K 4: 88K 5: 96K 6: 176K 7: 192K